DATA 202 — Quiz 4

Week 9 — Covering Weeks 6–7: kNN, Metrics, Trees & Cross-Validation

Name: _____ Date: _____

The Dataset: Campus Library

The campus library keeps four related tables (simulated data). Every question on this quiz uses them.

- `loans` — one row per book loan: `loan_id`, the `student_id` who borrowed it, the `book_id` borrowed, and `late` (**the label**: 1 = returned late, 0 = on time). There are 800 loans.
- `students` — one row per student: `student_id`, `class_year` (1–4) and `prior_late` (how many books they returned late last year).
- `books` — one row per book: `book_id`, `genre` and `pages`.
- `monthly_loans` — one row per student: how many books they borrowed in each month (`sep`, `oct`, `nov`).

The models in Questions 1–6 predict `late` from `pages`, `prior_late` and `class_year` (found by joining `loans` to `students` and `books`).

A few rows of `loans`:

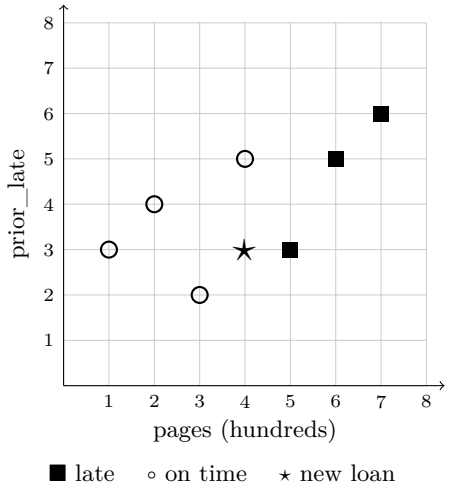
loan_id	student_id	book_id	late
L101	S011	B204	0
L102	S012	B087	1
L103	S012	B150	0
L104	S015	B204	1

A few rows of `students`:

student_id	class_year	prior_late
S011	2	0
S012	4	3
S013	1	1

Question 1 (SLO 06A)

Seven training loans are plotted by `pages` (in hundreds) and the student's `prior_late`: a **filled square** was returned **late**, an **open circle** on time. The **star** is a new loan. Use the straight-line distance (count the grid squares; you don't need the formula).



kNN with k = 1 predicts:

_____

kNN with k = 5 predicts:

_____

Which of the two k values is more likely to **overfit**? Why?

_____

_____

Question 2 (SLO 06B)

A kNN model (k = 5) is scored on the **200 test loans**. Its confusion matrix, with **late** as the positive class:

	predicted on time	**predicted late**
actually on time	132	15
actually late	31	22

Metric	Show the fraction	Value
Accuracy		
Precision (for late)		
Recall (for late)		
Accuracy of the lazy model that always says on time		

Question 3 (SLO 06C)

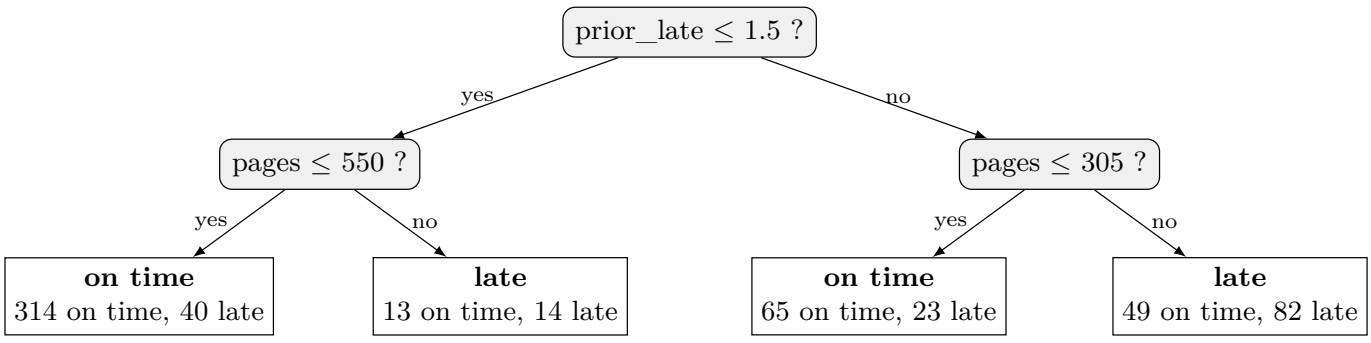
`data` holds all 800 loans. Fill in the blanks so that the score estimates how well the model will do on **new** loans. Then say how many loans end up in the test set.

```
X = data[["pages", "prior_late", "class_year"]]
y = data["late"]
X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.25, random_state=1)
scaler = StandardScaler().fit(_____________)
knn = KNeighborsClassifier(n_neighbors=5).fit(scaler.transform(_____________), _____________)
knn.score(scaler.transform(_____________), _____________) # accuracy on NEW loans
```

Number of loans in the test set: _____

Question 4 (SLO 07A)

A decision tree (`max_depth=2`) was fit on the 600 training loans. Each leaf shows its prediction and the training loans that reached it. Follow the tree for each new loan below.



prior_late	pages	class_year	Prediction (late / on time)
0	600	2	
3	200	1	
2	400	4	

Question 5 (SLO 07B)

Trees of different depths were scored with **5-fold cross-validation on the 600 training loans** (the test set was not used). The table shows the mean accuracy on the folds each tree was fit on, and on the held-out folds.

max_depth	1	2	3	4	6	8	no limit
training folds	0.735	0.795	0.802	0.820	0.856	0.897	0.935
5-fold CV (held-out folds)	0.718	0.770	0.765	0.758	0.747	0.728	0.710

Which `max_depth` would you choose, and why? Name one depth that **underfits** and one that **overfits**, using both rows.

Question 6 (SLO 07C)

A grid search tunes two hyperparameters of the tree:

```
grid = {"max_depth": [2, 4, 6, None],
    "min_samples_leaf": [1, 10, 30]}
search = GridSearchCV(DecisionTreeClassifier(), grid, cv=5)
search.fit(X_train, y_train)
search.best_params_ # {'max_depth': 4, 'min_samples_leaf': 30}
search.best_score_ # 0.782
search.score(X_test, y_test) # 0.74
```

How many hyperparameter **combinations** does it try, and how many **trees** does it fit during cross-validation? Then: why is the test score lower than `best_score_`?

Question 7 (SLO 04A) — Retake from Quiz 3

For each column below, say whether it is a **primary key**, a **foreign key**, or **neither** in that table. For a foreign key, name the table it points to.

Table	Column	Primary key / Foreign key / Neither	If foreign key: points to which table?
loans	loan_id		
loans	student_id		
students	student_id		
loans	book_id		
books	pages		

Question 8 (SLO 04B) — Retake from Quiz 3

Suppose `loans` and `students` contained **only the rows shown on page 1**. Each table below is the result of `pd.merge(loans, students, on="student_id", how=___)` (only three columns shown). Write which join made each one: **inner**, **left**, **right** or **outer**.

Result 1

loan_id	student_id	class_year
L101	S011	2
L102	S012	4
L103	S012	4
NaN	S013	1

how = _____

Result 2

loan_id	student_id	class_year
L101	S011	2
L102	S012	4
L103	S012	4

how = _____

Result 3

loan_id	student_id	class_year
L101	S011	2
L102	S012	4
L103	S012	4
NaN	S013	1
L104	S015	NaN

how = _____

Result 4

loan_id	student_id	class_year
L101	S011	2
L102	S012	4
L103	S012	4
L104	S015	NaN

how = _____

Question 9 (SLO 04C) — Retake from Quiz 3

Here are two rows of the wide table `monthly_loans`. Sketch the part of `loans_long` that comes from these two rows: column names and every row.

student_id	sep	oct	nov
S003	0	2	1
S007	2	0	2

```
loans_long = pd.melt(
   monthly_loans,
   id_vars=["student_id"],
   value_vars=["sep", "oct", "nov"],
   var_name="month",
   value_name="n_loans",
)
```